Yu-Han Huang

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EDUCATION

B.S. in Electrical Engineering, National Taiwan University (NTU), Taipei, Taiwan — Sep 2023–June 2027 (expected)

- Overall GPA: 4.04/4.30 (3.91/4.0 converted)
- Selected Coursework: Machine Learning, Deep Learning for Computer Vision, Deep Reinforcement Learning
- Research Interests: Speech and Audio Processing, GUI Agents, Vision-Language-Action Models

RESEARCH EXPERIENCE

Natural Language Processing Lab (NLP), UCLA — July 2026–Present

Visiting Researcher | Supervisor: Prof. Kai-Wei Chang

- Investigated GUI agents for multi-step web navigation and computer-use tasks
- Developed realistic tasks that required GUI agents to combine deep research with multi-step browser interactions

Speech Processing and Machine Learning Lab, NTU — July 2025–Present

Undergraduate Researcher | Advisor: Prof. Hung-Yi Lee

- Explored training-free approaches to improve large audio-language model understanding and response reliability
- Investigated safety and alignment methods for full-duplex spoken language models

Vision and Learning Lab, NTU — Sep 2025–June 2026

Undergraduate Researcher | Advisor: Prof. Yu-Chiang Frank Wang

- Studied vision-language-action models for long-horizon embodied task planning

INTERNSHIP EXPERIENCE

Vision Machine Learning Team, Synology Inc., New Taipei City, Taiwan — Nov 2026–Present

Research and Development Intern

- Trained video deep learning models to detect human falls in camera footage for safety monitoring

Open Cloud Infrastructure Software Center (OCIS), ASUS — May 2026–Nov 2026

AI Research Intern

- Conducted research on capability enhancement, evaluation, safety, and alignment for large audio-language models, full-duplex speech models, and LLM-based agents

Institute of Information Science, Academia Sinica — July 2024–Aug 2024

Student Researcher | Supervisor: Prof. Yennun Huang

- Developed automated pipelines for generating both attack and benign audit logs for intrusion detection research

PUBLICATIONS

* *Equal contribution*

[1] Yu-Han Huang, Chih-Kai Yang*, Ke-Han Lu*, An-Yu Cheng*, and Hung-yi Lee, "Enhancing How Large Audio-Language Models Hear: Identifying and Amplifying Acoustic Neurons in the Audio Encoder." **[arXiv:2607.11801]**. Accepted to the *2026 IEEE Spoken Language Technology Workshop (SLT)*.

[2] Kai-Po Chang, Chang-Hsun Wu, Yen-Siang Wu, **Yu-Han Huang**, Kai-Chen Yang, and Yu-Chiang Frank Wang, "Self-Correcting Imaginative Planning for VLM Embodied Agents," submitted to the *International Conference on Learning Representations (ICLR 2027)* (under review).

[3] Lok-Lam Ieong*, Chia-Chien Chen*, Chih-Kai Yang*, **Yu-Han Huang**, An-Yu Cheng, and Hung-yi Lee, "Nudging Hidden States: Training-Free Model Steering for Chain-of-Thought Reasoning in Large Audio-Language Models," in *Proc. Interspeech 2026*, 2026, pp. 3797–3802, doi: 10.21437/Interspeech.2026-554.

[4] Kai-Wei Chang, Yi-Cheng Lin, Huang-Cheng Chou, Wenze Ren, **Yu-Han Huang**, Yun-Shao Tsai, Chien-Cheng Chen, Yu Tsao, Yuan-Fu Liao, Shrikanth Narayanan, James Glass, and Hung-yi Lee, "TaigiSpeech: A Low-Resource Real-World Speech Intent Dataset with Scalable Data Mining In-the-Wild," in *Proc. Interspeech 2026* (Long Paper Track), 2026, pp. 7304–7313, doi: 10.21437/Interspeech.2026-1511.

[5] Ke-Han Lu et al., including **Yu-Han Huang**, "How Auditory Knowledge in LLM Backbones Shapes Audio Language Models: A Holistic Evaluation." **[arXiv:2603.19195]**. Accepted to the *2026 IEEE Spoken Language Technology Workshop (SLT)*.

[6] Yi-Cheng Lin et al., including **Yu-Han Huang**, "Findings of the First Teaching Monster Challenge: A Benchmark of Pedagogical Content Knowledge in AI Agents." **[arXiv:2608.08852]**, submitted to the *AAAI 2027 Artificial Intelligence for Social Impact Track* (under review).

TEACHING EXPERIENCES

Deep Learning for Human Language Processing, EECS Department, NTU Spring 2026

Teaching Assistant | Instructor: Prof. Hung-Yi Lee

- Co-organized the First Teaching Monster Challenge and co-developed its baseline pipeline
- Conducted human evaluation of AI-generated teaching videos and coauthored the paper

Machine Learning, EECS Department, NTU Spring 2026

Teaching Assistant | Instructor: Prof. Hung-Yi Lee

- Designed and prepared the course assignment on model merging

AWARDS & SCHOLARSHIPS

- **AI Talent+ Overseas Research Scholarship**, NTU July 2026–Aug 2026
  Funded visiting research at the UCLA NLP Lab (Prof. Kai-Wei Chang)
- **MakeNTU Hackathon – 1st Place, Enterprise Award**, Taipei, Taiwan May 2025
  AI Driving Coach: real-time driver analysis and voice feedback with the Google Gemini API and WebSocket streaming on a hardware-software driving simulator

COURSE PROJECTS

Drifting Q-Learning with Distributional Behavior Cloning Spring 2026

Deep Reinforcement Learning

- Co-developed DriftQL, an offline reinforcement learning method that replaces pointwise behavior cloning with a distributional drifting objective
- Conducted experiments against 13 baselines on 10 OGBench tasks, achieving state-of-the-art performance on three tasks with one-step inference

Detection-Guided Image Layer Extraction Fall 2025

Deep Learning for Computer Vision

- Built a two-stage image-decomposition pipeline using Qwen-Image-Layered for RGBA layer generation
- Refined layer boundaries with DocLayout-YOLO-based detection, reducing cross-layer content leakage

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

General Coordinator, NTUEE Night, NTU March 2026

- Led the department's largest annual event with 200+ performers and staff, attracting an audience of 800+

Director of Public Relations, NTUEE Student Association Fall 2025–Spring 2026

- Led the Public Relations Department and managed vendor outreach and partnerships for student association events

Vice Captain, NTUEE Women's Basketball Team Fall 2024

SKILLS

Programming & frameworks: Python, C/C++, Go, PyTorch, RISC-V, MATLAB

Tools & platforms: Git, Docker, Raspberry Pi, HTML/CSS

Languages: Mandarin Chinese (native), English (TOEFL iBT 5.5/6, 110/120)